Using PINNs for approximating the behavior of Damped and Ideal Harmonic Oscillators

David Fetterman
Northwestern University
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Motivation

- Harmonic Oscillators
  - Fundamental systems in physics
- Why PINNs?
  - Labeled data not required
- Goal: Implement a PINN to approximate the dynamics of Ideal and Damped Oscillator and compare to true solution found via Finite Distance Method (FDM)

Systems Studied

- Ideal Oscillator(undamped): $x'' + \omega_0^2 x = 0$
- Damped Oscillator: $x'' + 2\omega_0\gamma x' + \omega_0^2 x = 0$
- Initial Conditions x(0) = 1, x'(0)=v=0

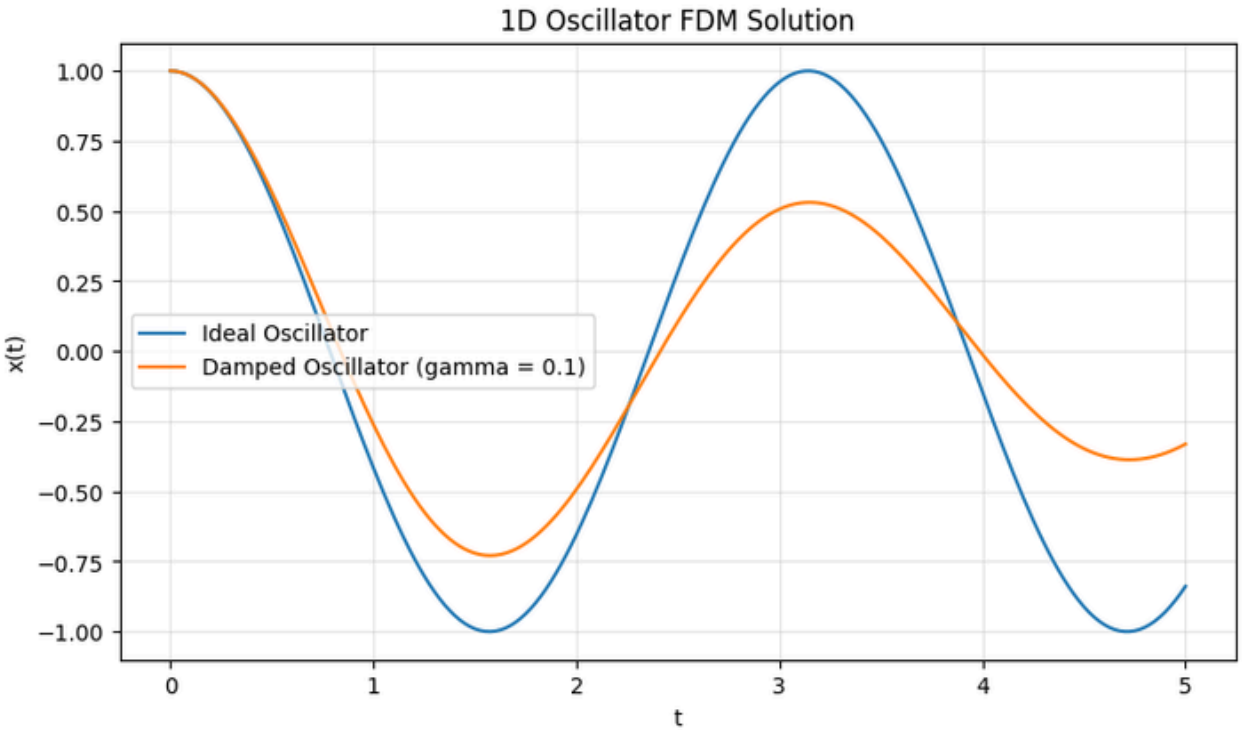
Analytical Solution (ideal): $x(t) = cos(\omega t)$

Analytical Solution (damped): $x(t) = e^{-\gamma t/2}\cos(\omega_d t)$

$$\omega_d = \sqrt{\omega^2 - \frac{\gamma^2}{4}}$$

Finite Difference Method (FDM)

- Generates true solution for training comparison
  - Central difference scheme used
  - Sparse noisy data sets were sampled from solution for PINN data loss function



PINN Architecture

- Fully connected NN
- Input: (t)
- Two hidden layers, 64 neurons each, tanh activation function
- Output: x(t)
- Xavier Initialization was used for stability
- Automatic differentiation was used to compute x’(t) and x’’(t)

PINN Loss Function

$$\mathcal{L}_{total} = \mathcal{L}_{PDE} w_{PDE} + \mathcal{L}_{IC} w_{IC} + \mathcal{L}_{BC} w_{BC} + \mathcal{L}_{Data} w_{Data}$$

$$\mathcal{L} = \frac{1}{N_f} \sum w_{\mathrm{PDE}} f(t)^2 + w_{\mathrm{IC}} \left[(x(0) - x_0)^2 + (x'(0) - v_0)^2 \right] + w_{\mathrm{Data}} \frac{1}{N_d} \sum (x_{\mathrm{pred}}(t_i) - x_{\mathrm{data}}(t_i))^2 + w_{\mathrm{BC}} L_{\mathrm{BC}}$$

Respective weights were defined as:

$w_{PDE} = 1.0$

$w_{IC} = 5.0$

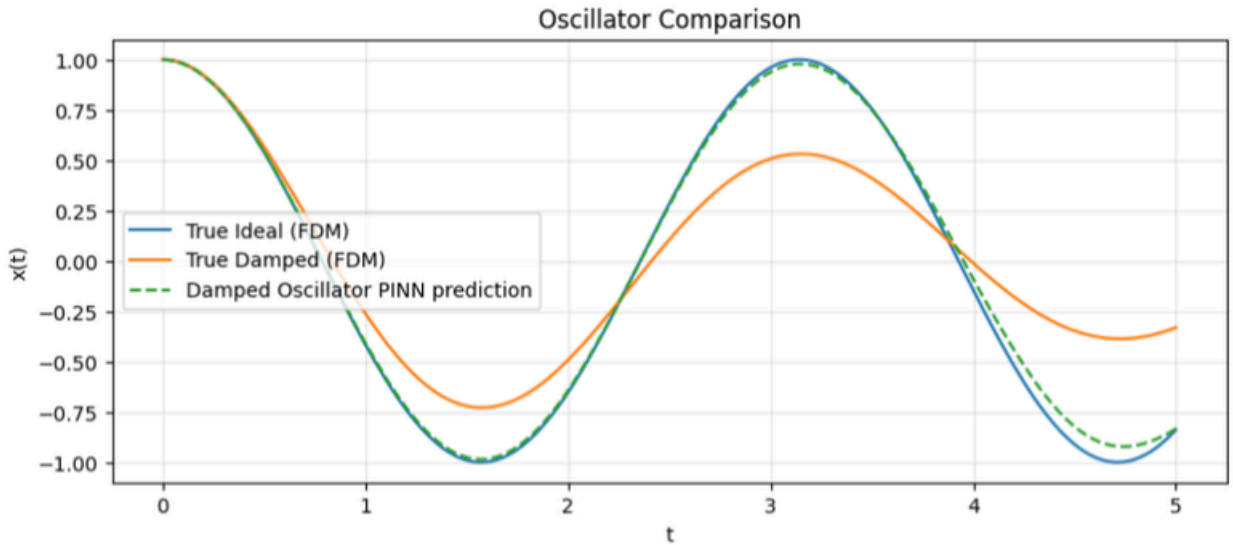
$w_{BC} = 1.9$

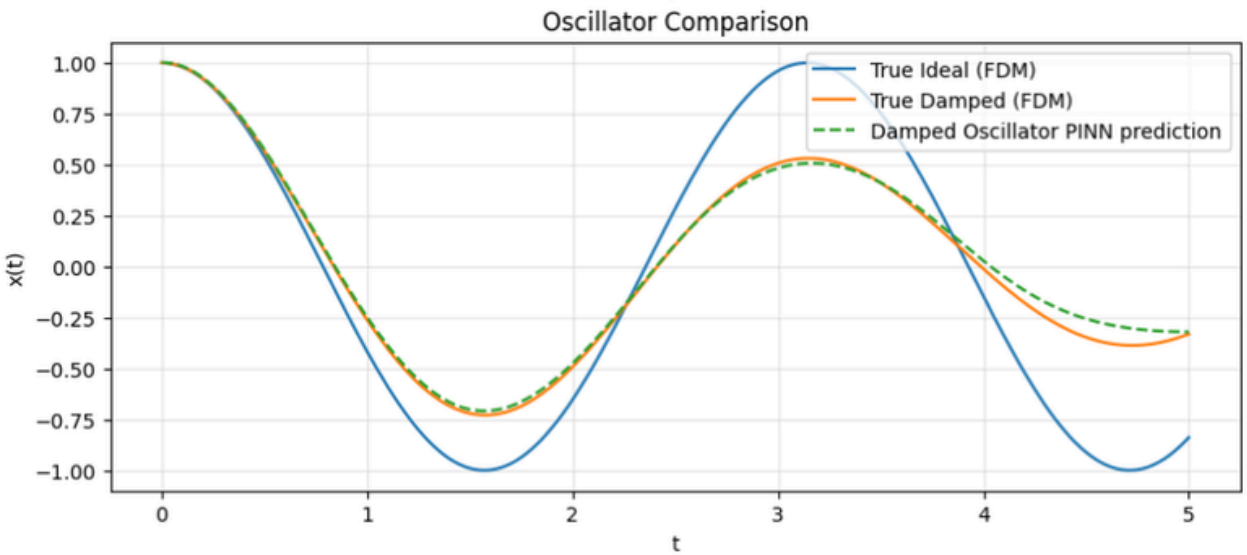
$w_{Data} = 10.0$

Training Procedure

- Optimizer: ADAM, Learning Rate: 10^-3
- Iterations: 1500
- Random sample of collocation, initial, and data points each iteration

PINN Predictions





Discussion

- PINNs can effectively learn solutions without fully labeled datasets
  - Comparison of PINNs predicted solution to real
  - Not just for damped and ideal oscillators but works for many systems.
- Future Directions
  - Explore more complex systems including ones with no closed form solution such as chaotic systems
  - Include systems that have boundary conditions involved.